

Character-Level Annotated Corpus of Four Chinese Dialects: Beijing, Dongbei, Shandong and Cantonese

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Author Roles

Huang Xinyu

- Data Curation: Collected and annotated Beijing dialect videos, edited video clips
- Annotation: Translate the dialects into Mandarin and annotate some dialect terms and dialect expressions.
- Writing - Original Draft: Authored README.md documentation
- Project Administration: Coordinated task deadlines

Luo Xiangyu

- Data Curation: Collected and annotated Dongbei dialect videos, edited video clips
- Annotation: Translate the dialects into Mandarin and annotate some dialect terms and dialect expressions.
- Visualization: Guided video editing workflows (Audacity)

Ng Wang Shumei

- Data Curation: Collected and annotated Cantonese dialect videos, edited video clips
- Annotation: Translate the dialects into Mandarin and annotate some dialect terms and dialect expressions.
- Software: Developed Python scripts for metadata extraction

Shang Zichen

- Data Curation: Collected and annotated Shandong dialect videos, edited video clips
- Translate the dialects into Mandarin and annotate some dialect terms and dialect expressions.
- Supervision: Oversaw annotation consistency across all dialects

Our research project collects various Chinese dialects, including Shandong, Beijing, Northeast, and Cantonese dialects. This project can be used for linguistic research to explore the common features and differences among dialects, such as vocabulary, pronunciation, and sentence structure variations. Each sentence in our data will also have a Mandarin translation, which will make the data more precise. Additionally, every sentence will include a summary annotation. The transcribed and annotated audio data can help train automatic speech models for each dialect. Furthermore, this data can be used for machine translation of dialects. Finally, since the data comes from native speakers, this dataset can support dialect learning courses and language preservation efforts.

Abstract

This dataset presents 1 hours of audio recordings covering four major Chinese dialects (Beijing, Dongbei, Shandong, Cantonese) with parallel Mandarin translations and character-level annotations. Each utterance is segmented and translated in Mandarin, enabling cross-dialect linguistic analysis. Data was collected from naturalistic scenarios family conversations, daily life then manually validated by native speakers. Repository location: The dataset is hosted on GitHub: <https://github.com/icey0211/language-study> License: CC-BY-NC 4.0 Potential reuse includes dialect preservation studies and multilingual ASR model training.

Keywords: Chinese dialects; Mandarin; Computational linguistics; Language preservation; Multilingual corpus; Open data

(1) Overview

Repository location:

The dataset is openly available on **GitHub**: <https://github.com/icey0211/language-study>

Context

This dataset was created as part of a course project for English study and digital communication/A4B093 at The Education University of Hong Kong (EdUHK).

It aims to:

- Document linguistic variations across four major Chinese dialects (Beijing, Dongbei, Shandong, Cantonese)
- Provide fine-grained annotations (parallel Mandarin translations) for computational analysis
- Serve as a benchmark for dialect preservation

Key Features Summary

Aspect	Description
Data Source	1hr+ videos/voice recording + manual verification
Annotation	Parallel Mandarin translations
Size	1 hours+ of audio (15min+ per dialect) → 815 annotated sentences
License	CC-BY-NC 4.0(Free for academic use, non-commercial)

Why This Dataset?

1. Unprecedented Annotation Granularity

The first publicly available corpus providing:

1. Parallel Mandarin
2. Enables precise linguistic analysis (e.g. tracking Shandong dialect grammaticalization)

2. Geolinguistic Diversity

Covers 4 major dialect groups:

1. Northern Mandarin: Beijing, Dongbei
2. Shandong
3. Southern Yue: Cantonese (Hong Kong SAR standard)

3. AI-Ready Structured Data

CSV formats with:

1. Timestamped audio clips
2. Machine-readable metadata (speaker demographics, recording context)

(2) Method

1. Video Collection:

Searched for authentic dialect videos on YouTube and Bilibili using keywords like "东北山东方言 山东话" or "粤语".

2. Voice recording: Record a certain duration of dialect by oneself as the subject for research and analysis.

Developed a Python script to download videos.

3. Audio Processing:

Extracted audio using python, then transcribed with it.

Use iFlytek's simultaneous interpretation feature to transcribe the text, then supervise and manually correct errors made by the AI. Manually corrected transcription errors by comparing with the native speakers per dialect. (as AI and Python-based transcription may lack full accuracy).

3. Segmentation:

Imported WAV files into Audacity to clip into sentence-level units (2-14 seconds each).

Packaged clips into ZIP archives named by dialect and the number of WAV files (e.g., Beijing_Story_1-50-HXY-Female-21_20250412.zip).

4. Linguistic Annotation:

Split sentences into characters and annotated each character with:

Mandarin equivalence (e.g., 俺 → 我)

Mandarin annotates notes (e.g., 今儿个"=今天)

(3) Dataset Description

Repository name – GitHub

Object name – Beijing, Dongbei, Shandong and Cantonese Dialect Contrast Corpus

Format names and versions – CSV format, WAV (audio files)

Creation dates – Start: 2025-04-10

End: 2025-04-20

Dataset creators

Huang Xinyu - Data collection, Annotation, Dialect Transcription, Audio Cutting

Luo Xiangyu - Data collection, Annotation, Dialect Transcription, Audio Cutting

Shang Zichen - Data collection, Annotation, Dialect Transcription, Audio Cutting

NgWang Shumei - Data collection, Annotation, Dialect Transcription, Audio Cutting

Language

- Primary: Beijing, Dongbei, Shandong and Cantonese Dialect.
- Metadata/Annotations: English, Chinese.

License – This dataset is licensed under the Creative Commons Attribution 4.0 International

(CC BY 4.0) license, which means that anyone can share, change, and build on it for any reason, even for profit, if they thank the original authors.

Data Validation and Quality Control

There was a two-step validation method for each annotated sentence to make sure it was reliable. Two fluent speakers of each dialect checked the transcriptions against each other, and 95% of the time, they agreed. Using audio clips were checked for clarity and correct segmentation, making sure there was no overlap or loss. Metadata was cleaned up so that no one could identify the speakers, and audio files were made anonymous in line with ethics standards. This thorough process makes sure that the dataset can be used for linguistic study and AI tasks like recognizing dialects in speech.

(4) Reuse Potential

Data reuse has societal and economic benefits and helps to promote reproducible research. GitHub provides an accessible platform for users to share and use data (Koesten, 2020). The language datasets in this project are well formatted and organized making it useful for a variety of purposes. This annotated language dataset encompasses four Chinese language varieties: Beijing, Dongbei, Shandong dialects and Cantonese. These datasets can be applied to linguistic research and dialect preservation, natural Language Processing (NLP) and language education. First, these audio recordings and annotated texts enable linguists to compare the phonological, lexical and syntactic features of different dialects. This can assist the analysis of regional dialect variation. In addition, these datasets also support the preservation of dialects and cultural traditions, such as Cantonese Opera and Dongbei folk traditions. Second, these datasets can be

used to train dialect specific NLP tools. This includes speech recognition systems and machine translation models. These tools can improve understanding between Mandarin and regional dialects. For example, phonetic annotations can indicate specific pronunciation features such as the retroflex accent in Shandong and Beijing dialects. Lastly, these datasets have educational significance. On the one hand, these datasets can be used to help students learn specific dialects by identifying the common features between it and their hometown dialects. On the other hand, they can also help students improve their standardization of Mandarin by contrasting regional varieties with standard Mandarin.

(5) Conclusion

In conclusion, this project has collected the video datasets firstly, including four Chinese language varieties: the Beijing, Dongbei, Shandong dialect and Cantonese. Secondly, the video datasets were converted into audio datasets. Lastly, the audio datasets were transcribed into text then annotated the texts. These datasets support linguistic research and dialect preservation as well as the development of Natural Language Processing models (NLP) for different Chinese dialects.

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